CIS 4930 — Final project

REPO LINK: CIS_4930_Final_Group_Project

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Architecture

Flask was used for the backend of our project. Some endpoints used:

API

- `GET /api/tasks` — list tasks as JSON
- `POST /api/tasks` — send JSON: `name`, `start_date`, `end_date`, optional `notes`. Server adds `id`.
- `GET /health` — returns `{"status":"ok"}`

Bad requests get status `400` and `{"error":"..."}`.

Dates are plain strings (ISO-style is fine, e.g. `2026-05-01`).

HTML/CSS/JS was used for our frontend, and the code in the frontend connected to app.py.

Tools used

- Git/GitHub
- Docker/Docker compose
- Jenkins
- Flask app for tasks (name, start/end dates, notes). Tasks are stored in memory only, so they disappear when the server restarts.

Setup

Jenkins was downloaded and tested on all group members' machines. We created a new item and used a pipeline to connect to this repository.
Once Jenkins is fully linked with our repository, it is now ready to automate the workflow of our project.
Once Jenkins runs our project, we can access it
with http://localhost:5000

Workflow

1. **Trigger** — A Jenkins build starts the pipeline.

2. **Checkout** — Jenkins copies the repository into the workspace.

3. **Build** — `docker compose build` uses the `Dockerfile` to install dependencies and produce a container image of the application.

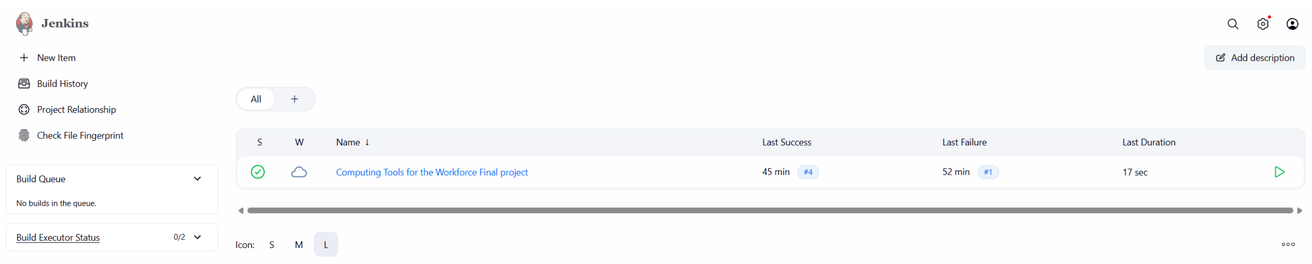
4. **Deploy** — `docker compose up -d` runs the container in the background and maps host port **5000** to the app.

5. **Verify** — After a short wait, the pipeline requests `http://localhost:5000/`. A successful response passes the build; a failure fails the build.

Summary: The pipeline automates *checkout* → *build image* → *run container* → *smoke test* so each change is built and checked the same way.

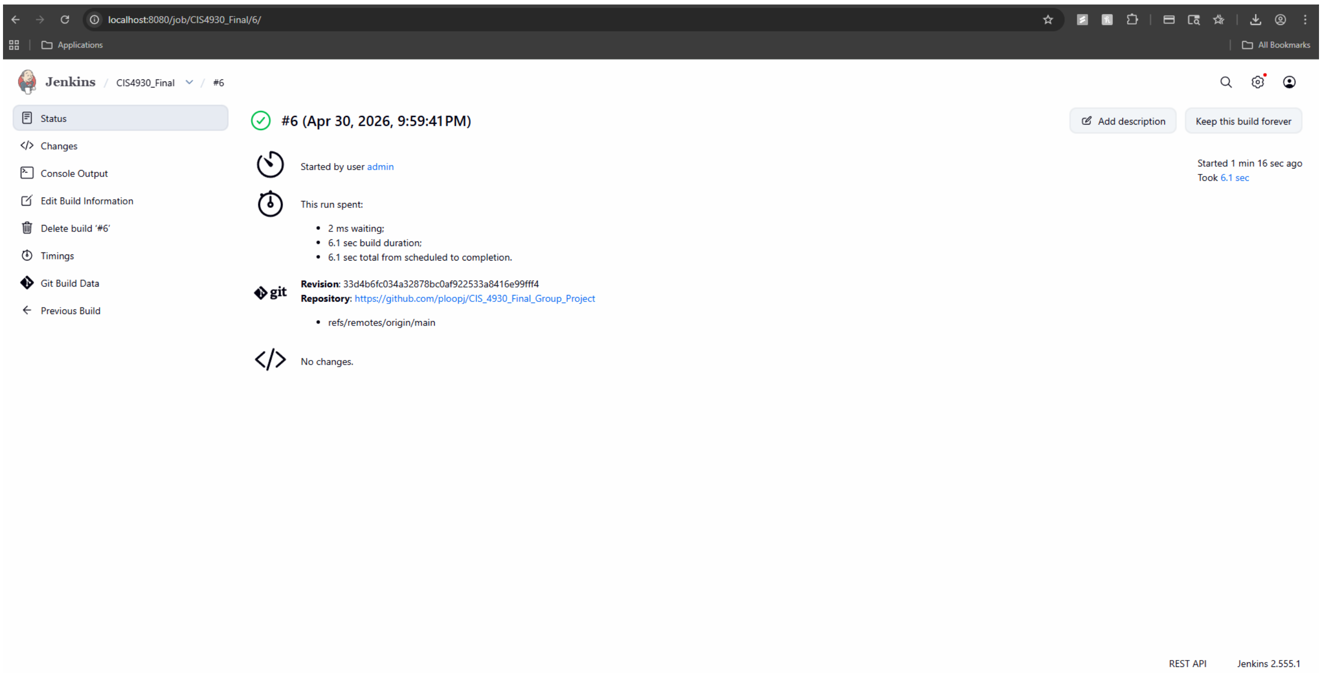
Screenshots

Jenkins Home Page



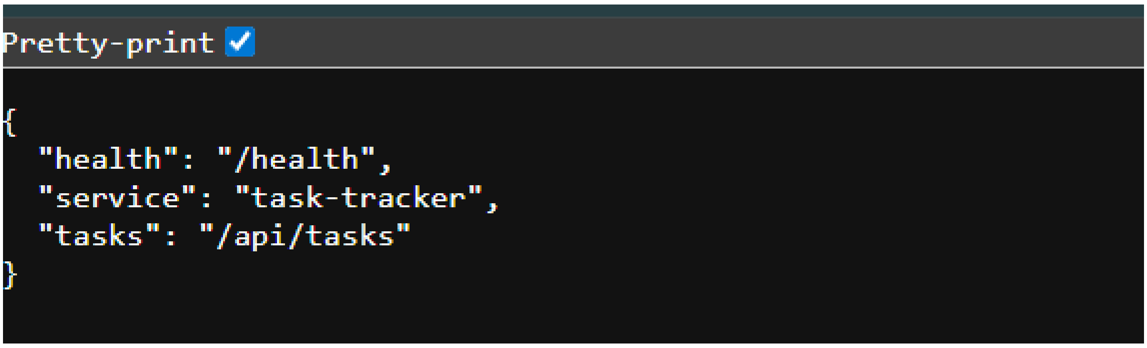
Jenkins home page that lists our project.

Jenkins Build Page



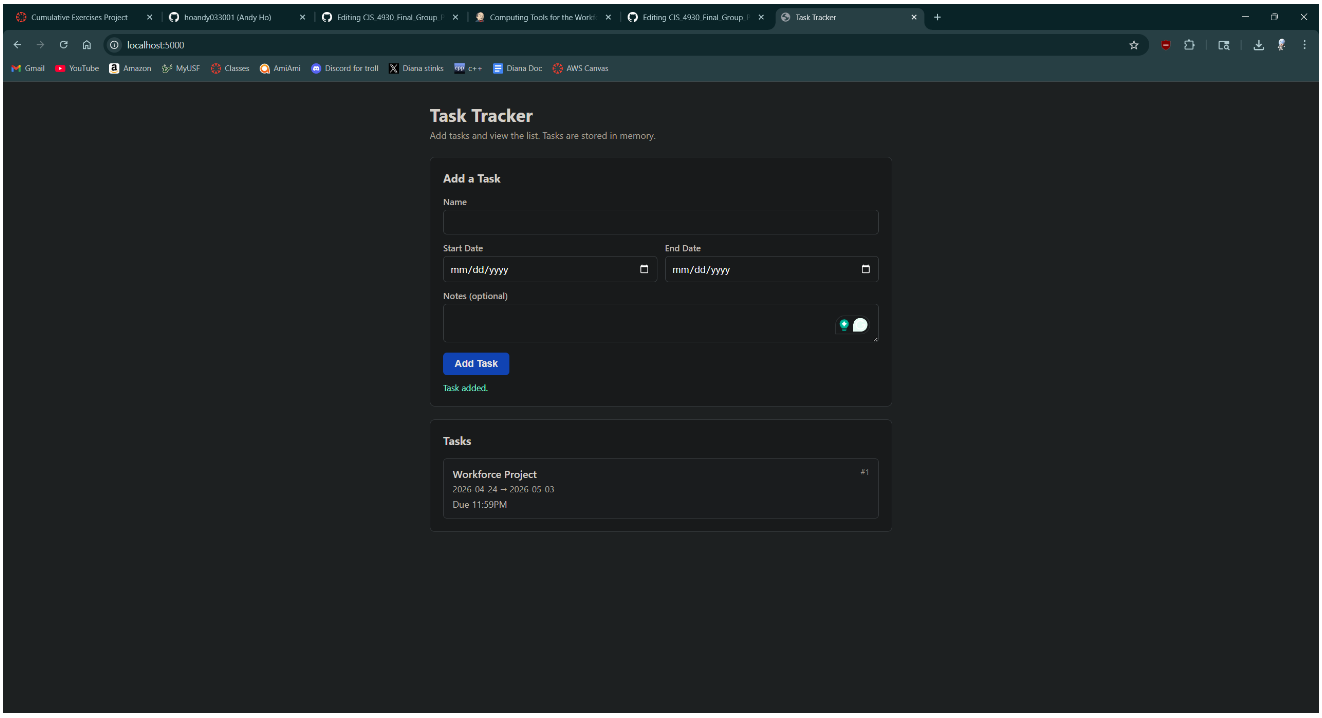
Our Jenkins configuration is successfully running with our Jenkinsfile setup.

Skeleton Page



This was our skeleton website that ran with all of our files. We will build off of this

Task Tracker Project



This is our finalized project. It is a simple task tracker that lets you create a list of tasks to stay organized. Jenkins automates the process of running our code with the help of Docker.